

National University of Computer and Emerging Sciences

Q1)

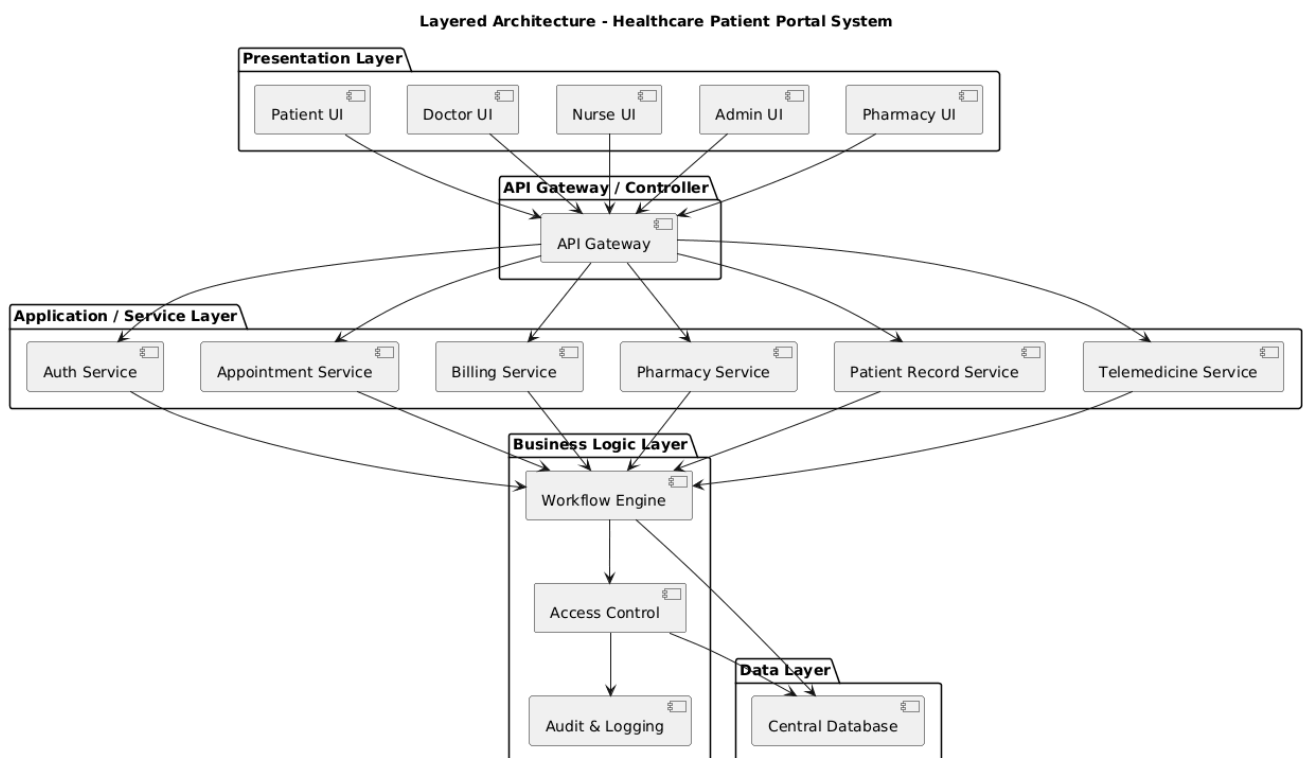
Tasks:

- Identify the most appropriate Architectural Pattern. Justify your answer(4-pts).
- Using Box and line, draw the architecture diagram that clearly shows modules, subsystems and control flow. Incorporate post deployment changes in the architecture diagram as well. (4-pts)

The most appropriate architectural pattern for the CHPPS is a combination of Layered Architecture with Microservices (or Service-Oriented Architecture).

Justification:

- Layered Architecture is suitable because:
 - It separates the system into Presentation, Business Logic, and Data layers, supporting role-based interfaces (patients, doctors, admin).
 - Ensures centralized data consistency since all layers interact with a unified data source.



Q2:

- a. Refer to Figure 1 given at the end, do the following:
- i. Calculate the Cyclomatic Complexity (CC). **[1-pts]**

Handwritten calculation of Cyclomatic Complexity (CC):

$$CC = E - N + 2P$$
$$= 10 - 15 + 2$$
$$CC = 6$$

- ii. Identify all linearly independent paths. **[2-pts]**

Handwritten list of linearly independent paths:

- ① A B D E J End.
- ② A B D E H J L M End.
- ③ A B D E H K L M End.
- ④ A B C F H J L M End.
- ⑤ A B C G H K L M End.
- ⑥ A B C ---- H L M N back
H m(F) End.

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- b. Consider a program named 'Multiply' that takes two integer inputs x and y and produces the product of x and y as output. The valid input ranges are defined as follows: $10 \leq x \leq 50$ and $5 \leq y \leq 25$. Perform boundary values analysis (BVA) including Nominal value for both input variables x and y.

- I. Fill the table below for each variable. Identify Min, Min+1, Nominal (Mid), Max-1, Max values. **[2-pts]**

Variable	Min	Min+1	Nominal (Mid)	Max-1	Max
x	10	11	30	49	50
y	5	6	15	24	25

- II. Calculate Generate ten Boundary Test Points basis following rule: **[2-pts]**

- Vary one variable at a time
- Keep the other variable fixed at its nominal value
- For each value of x: y = nominal , For each value of y: x = nominal

For x (y = 15 fixed)

(30,5), (30,6), (30,15), (30,24), (30,25)

For y (x = 30 fixed)

(10,15), (11,15), (30,15), (49,15), (50,15)

- III. Calculate Total no of Test Cases ensuring efficient coverage of boundary conditions. Table format: Test Case No., Test Case, Input (x, y), Expected O/P (x × y). **[3-pts]**

TC No.	Test Case Description	Input (x, y)	Expected Output (x × y)
TC1	x = Min	(10,15)	150
TC2	x = Min+1	(11,15)	165
TC3	x = Nominal	(30,15)	450
TC4	x = Max-1	(49,15)	735
TC5	x = Max	(50,15)	750
TC6	y = Min	(30,5)	150
TC7	y = Min+1	(30,6)	180
TC8	y = Nominal	(30,15)	450
TC9	y = Max-1	(30,24)	720
TC10	y = Max	(30,25)	750

Q3: Provide Suitable Reasoning

[7-Marks, 10 minutes]

1. **Based on UI Desing Concepts [3-pts]**

- The “Add Task” button is small and placed at the top of the right panel. Which UI principle is being violated? Suggest a redesign. **Fitts Law**
- Users cannot cancel tasks once they start editing, and the undo icon only works after task submission. Which UI principle is missing? **Controlled Autonomy**
- The colors scheme for high-priority tasks differs between project list and task list. Which principle is violated? How to fix it? **Consistency**

2. The principles of firmness, commodity, and delight. “Consider a mobile banking application that frequently crashes, has confusing navigation, but provides all required features. Analyze the system and identify which of the three principles are satisfied and which are violated. Justify your answer. [2-pts]

 Firmness:

It means the software should be **reliable and free from bugs**. The system must function correctly without crashes or errors that stop its operation.

 Commodity:

It refers to the **usefulness of the software**. The system should meet user requirements and be suitable for the purpose it was designed for.

 Delight:

It represents the **user experience**. The software should be easy to use, visually appealing, and provide a pleasant interaction to users.

3. A software product is highly functional and fast but difficult to modify and extend. Which FURPS attribute is weak? Enlist [2-pts]

The weak attribute is **Supportability**